

4-BIT ARITHMETIC LOGIC UNIT

MINI PROJECT OF EC4010



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INTRODUCTION

A 4-bit Arithmetic Logic Unit (ALU) is a digital circuit that performs a variety of arithmetic and logic operations on 4-bit binary numbers, making it an essential building block in microprocessors and digital systems. It processes data by taking two 4-bit inputs (operands), along with control signals that dictate the specific operation to be executed. Typical arithmetic functions include addition and subtraction, while logical functions may encompass AND, OR, XOR, and NOT operations. Some designs also support bit shifting and rotation.

The ALU generates a 4-bit output as the result of the operation, along with status flags such as carry, overflow, zero, or negative, which provide additional information about the outcome of the computation. These flags are crucial for higher-level decision-making processes in CPUs, such as branching and conditional execution.

Compact and efficient, 4-bit ALUs are widely used in simple processors, calculators, and embedded systems, serving as a foundational learning tool in digital electronics and computer engineering. They illustrate key principles of how larger and more complex processors operate by combining and scaling similar structures. Despite their simplicity, 4-bit ALUs demonstrate the power and versatility of digital logic in computational tasks.

TRUTH TABLE

The truth table for a 4-bit Arithmetic Logic Unit is as follows:

TABLE 01: TRUTH TABLE FOR 4 BIT ALU

A	B	Op	Operation	Result	Flags (Carry, Zero, Overflow)
0000	0000	00	Addition	0000	C=0, Z=1, O=0
0001	0001	00	Addition	0010	C=0, Z=0, O=0
0010	0001	00	Addition	0011	C=0, Z=0, O=0
1111	0001	00	Addition	0000 (with carry)	C=1, Z=1, O=0
0011	0001	01	Subtraction	0010	C=0, Z=0, O=0
0100	0011	01	Subtraction	0001	C=0, Z=0, O=0
0011	0011	01	Subtraction	0000	C=0, Z=1, O=0
0001	0001	10	Multiplication	0001	C=0, Z=0, O=0
0010	0010	10	Multiplication	0100	C=0, Z=0, O=0
0011	0010	10	Multiplication	0110	C=0, Z=0, O=0

CIRCUIT DIAGRAM

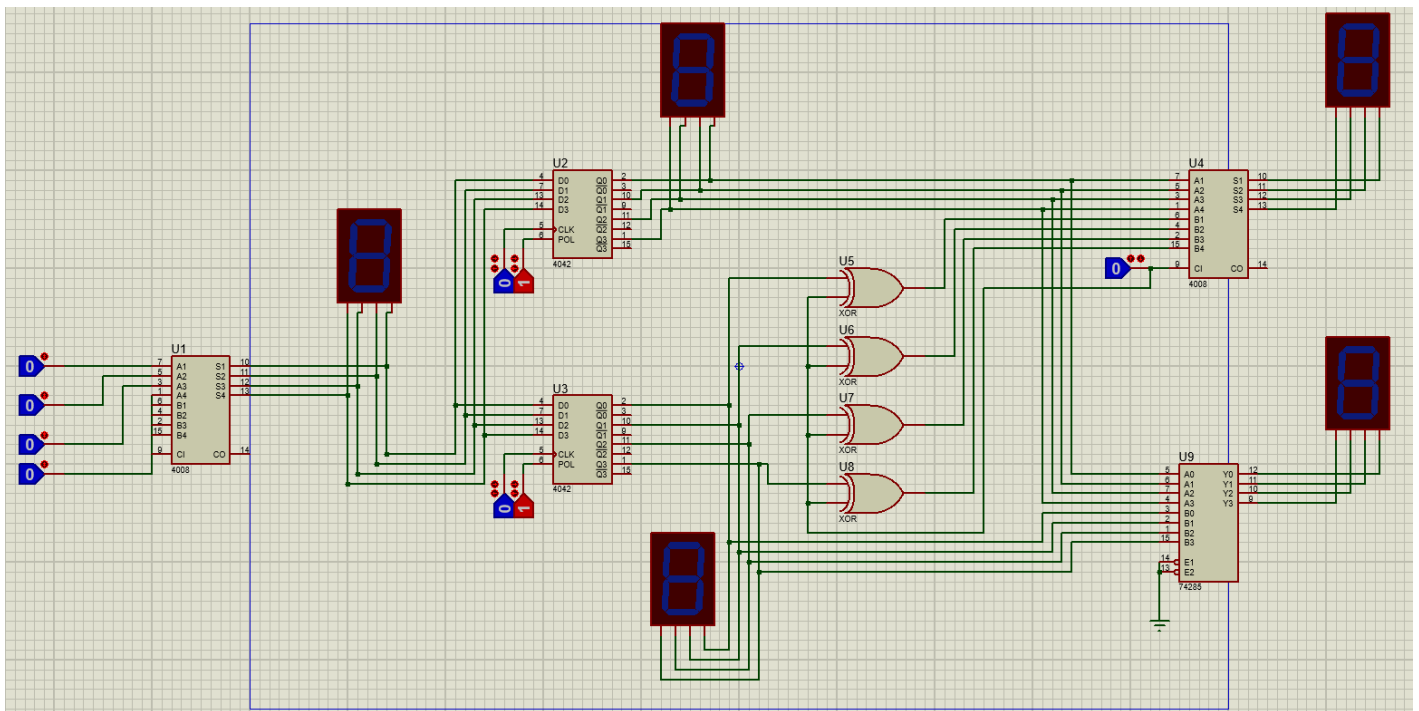


FIGURE 01 : CIRCUIT DIAGRAM FOR 4 BIT ALU

LIST OF COMPONENTS WITH BUDGET

COMPONENT	QUANTITY	UNIT PRICE (Rs.)	TOTAL PRICE(Rs.)
7 Segment Display	5	100	500
XOR Gate 7486	2	50	100
7 Segment Decoder	1	50	50
Bread Board	1	500	500
CD 4042-Quad Flip-Flop IC	2	300	600
CD 4008-4 Bit Binary Full Adder	1	300	300
SN 74285- 4Bit Parallel Binary Multipliers	1	400	400
Switches	10	20	200
MM74C922	1	500	500
TOTAL PRICE			3150

CONCLUSION

The design and implementation of the 4-bit Arithmetic Logic Unit (ALU) demonstrate the fundamental principles of digital computation and microprocessor functionality. This ALU supports three critical operations: addition, subtraction, and multiplication, making it a versatile yet compact computational unit suitable for educational and practical applications.

The addition operation is achieved using a ripple-carry adder circuit, ensuring accurate handling of binary sums, including carry propagation. Subtraction is implemented through a two's complement approach, where the binary inversion and addition of a carry input enable effective subtraction. Multiplication, a more complex operation, is managed using a parallel binary multiplier IC, showcasing the ALU's ability to handle higher-order operations.

Control signals allow seamless selection of these operations, while output flags such as carry, zero, and overflow provide essential status feedback. These features not only enhance the ALU's operational capability but also illustrate the importance of status flags in higher-level decision-making processes within CPUs.

Through this project, we have demonstrated the integration of various digital components, including logic gates, multiplexers, adders, and multipliers, to create a cohesive and functional unit. The detailed truth table and component analysis further emphasize the importance of planning and accuracy in digital system design.

This project not only reinforces theoretical knowledge of digital logic and arithmetic operations but also provides practical experience in circuit design and debugging. The successful realization of the ALU highlights the scalability of such designs, offering a foundation for understanding more complex processors and digital systems.

In summary, the 4-bit ALU serves as a testament to the power and efficiency of digital design, proving to be a valuable learning tool for future advancements in computational technology.